

# Hamiltonian Vision Kernel: Symmetry-Pooled Quantum Correlator Features with Hardware Validation

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## Abstract

We introduce the Hamiltonian Vision Kernel (HVK), a hybrid quantum–classical architecture for image reconstruction. HVK assembles matrix-product-state patch features, a variational quantum circuit exposing a Pauli-correlator latent space, two-dimensional Fourier positional encoding, a learnable Hamiltonian energy diagnostic, and a classical decoder into a single pipeline. Two instantiations, HVK1D and HVK2D, realise chain and grid interaction graphs. Group-averaged observable pooling makes the HVK2D feature map exactly equivariant to the eight symmetries of the square. The contribution is this specific assembly rather than any ingredient in isolation. Across six real image datasets, per-image-trained HVK2D models reconstruct with peak signal-to-noise ratio (PSNR) between 25.8 and 41.6 decibels. On a resource-matched, multi-seed held-out comparison using CIFAR-10, a pre-declared TOST equivalence procedure at a one-decibel margin establishes that HVK2D is statistically *competitive* with **a resource-matched raw/local linear classical control**. Five fitted checkpoints are replayed on superconducting quantum hardware without retraining, reaching 25.90 to 31.52 decibels. HVK adds an entangling observable channel that is representationally necessary on a leakage-audited task requiring distant-patch products, an interpretable Hamiltonian energy diagnostic, and native hardware execution. We make no claim of quantum advantage.

**Keywords:** Quantum image processing, Tensor networks, Variational quantum algorithms, Pauli observables, Group-equivariant representations, Quantum hardware validation

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## 1 Introduction

Hybrid quantum–classical models are now often proposed for vision tasks. The aim is to obtain representations with a favourable inductive bias. Such models can use entanglement, physically motivated regularisers, or tensor-network structure, which standard convolutional networks do not have (Cerezo et al. 2021; Tilly et al. 2022; Preskill 2018). The variational quantum eigensolver and its later variants use the fact that physical Hamiltonians split into local terms. This allows them to build shallow circuits suited to the problem (Cerezo et al. 2021; Tilly et al. 2022; Bauer et al. 2020). Quantum image processing gives several ways to store pixel values directly in qubits. These range from amplitude-style encodings (Le et al. 2011) to basis-encoded pixel values (Zhang et al. 2013). Basic image operations on these encodings have also been developed (Chakraborty et al. 2022; Zhou et al. 2017; Yao et al. 2017). Tensor-network methods work well as compressed representations of structured classical data,

such as images (Cirac et al. 2021; Latorre 2005; Han et al. 2018; Stoudenmire and Schwab 2016; Ran et al. 2020; Guala et al. 2023) and particle-physics events (Araz and Spannowsky 2022). Hybrid models have also been applied to related imaging tasks, such as tomographic reconstruction (Haga 2024; Nau et al. 2023) and ghost imaging (Zhai et al. 2025). Quantum convolutional and capsule networks are further examples (Slabbert and Petruccione 2024; Cong et al. 2019; Hur et al. 2022; Liu et al. 2023). HVK focuses on the combination of a Pauli-correlator latent space, enforced group-equivariant pooling, and a physically motivated energy diagnostic.

We introduce the Hamiltonian Vision Kernel (HVK1D/HVK2D), which combines these ingredients into a single architecture with three properties studied jointly. The first is a *Pauli-correlator latent space*, in which measured VQC observables rather than a raw quantum state form the representation handed to a classical decoder, following quantum-enhanced feature maps (Havlíček et al. 2019). The second is an *exact, numerically-verified  $D_4$ -equivariant observable-pooling construction*, which group-averages the observable grid over the eight symmetries of the square rather than motivating symmetry through architecture choices alone. This follows group-equivariant classical vision (Cohen and Welling 2016; Bronstein et al. 2021) and the equivariant-quantum-neural-network line that formalises symmetry-embedding constructions for variational circuits (Meyer et al. 2023; Nguyen et al. 2024; Schatzki et al. 2024; West et al. 2024, 2025). The third is a *learnable Hamiltonian energy diagnostic*, with full nearest-neighbour  $XX/YY/ZZ$  sectors in HVK1D and grid-edge  $ZZ$  terms in HVK2D. Table 9 contrasts these properties against classical, adversarial, and standard quantum autoencoder baselines.

## Novelty relative to the nearest hybrid-vision architectures

HVK’s closest architectural relatives are three hybrid quantum-vision patterns already established individually in the literature.

*Register-based quantum autoencoders* (Romero et al. 2017; Wang et al. 2024; Asaoka and Kudo 2025) encode an image’s pixel amplitudes directly into a qubit register and train a trash-qubit fidelity objective. HVK never amplitude-encodes pixels into qubits. It exposes only classically pre-compressed MPS patch features as circuit input, and hands the decoder measured Pauli correlators rather than register amplitudes or a fidelity score.

*Quantum-kernel vision models* (Havlíček et al. 2019; Beaulieu et al. 2022) embed an input into a quantum feature space and read out a scalar kernel value consumed by a classical kernel method. HVK reads out an explicit vector of local and pairwise Pauli expectation values as a latent representation for a trained decoder, and adds enforced group-equivariant pooling and a Hamiltonian regulariser that a kernel construction has no analogue for.

*Tensor-network (TN) circuit classifiers* (Guala et al. 2023; Stoudenmire and Schwab 2016; Han et al. 2018) use MPS structure either as a standalone classical model or as a template for a shallow quantum circuit trained for classification. HVK uses MPS compression only as a classical *preprocessing* step that sets a separately trained VQC’s rotation angles, and targets per-image reconstruction rather than classification.

HVK’s contribution is the assembly of these four elements in one pipeline (Table 9). Section 5.2 examines how the individual ingredients recur across this literature.

## Contributions

1. **Architecture.** HVK1D/HVK2D combines MPS patch features, a six-qubit VQC, a 19–27-dimensional Pauli-correlator vector, 2D Fourier positional encoding, a graph-Hamiltonian diagnostic, and a classical decoder.
2. **Reconstruction scope.** Identically configured per-image fitting experiments cover six real image datasets (Section 4.1).
3. **Competitive with resource-matched classical baselines.** A pre-declared TOST equivalence test on held-out CIFAR-10 shows HVK2D is statistically equivalent, within a  $\pm 1$  dB margin, to **five of seven distinct** resource-matched classical/ablated controls, and substantially outperforms the two null controls (strict classical random features and a random-VQC) (Section 4.3).
4. **Hardware pilot.** Five fitted checkpoints are replayed on IBM hardware without retraining and decoded from hardware-measured observables (Section 4.4), strengthened by a noise/shot-budget sweep and repeated execution across backends (Sections 4.5–4.6).
5. **Equivariant observable pooling.** An exact, numerically-verified  $D_4$ -equivariant observable-pooling construction makes the HVK2D observable map equivariant to the eight symmetries of the square, to floating-point precision (Section 4.7). The guarantee follows from Reynolds group averaging over the  $D_4$  orbit rather than from anything quantum. The identical construction applied to a purely classical feature map yields the same exactness (Supplementary Material). We therefore report it as a property the assembled pipeline provides, not as a quantum result.
6. **Restricted nonlocal result.** In a leakage-audited fixed-readout protocol, entangling pair observables are necessary when the target depends explicitly on distant-patch products (Section 4.8).

A companion document, *Supplementary Material for the Hamiltonian Vision Kernel: Resource-Matched Ablations, Validation, and Reproducibility*, reports resource-matched held-out comparisons of HVK and classical/ablated feature maps, together with full hardware-pilot methodology covering circuit construction, validation, job identifiers and quota accounting. It also gives provenance audits of excluded exploratory results, among them the training-dynamics traces of Section 4.9, which we report as an interpretability readout rather than as evidence of a transition. Section 4.3 summarises the central finding of the held-out comparison and how it relates to the results here.

## 2 Hamiltonian Vision Kernel: Architecture

HVK treats each spatial patch as a finite spin state and compresses it as a matrix product state (MPS). A variational quantum circuit (VQC) then measures a fixed set of Pauli observables, which a classical multi-layer perceptron (MLP) decodes back into

pixels. A learnable graph-Hamiltonian energy term regularises training. Parameterised Hamiltonian learning approaches treat the Hamiltonian as the primary learned object (Shi et al. 2023); in HVK it is instead a diagnostic regulariser alongside a separate reconstruction objective.

### ***Problem statement.***

Given an image  $x$  partitioned into non-overlapping patches  $\{x_p\}_{p=1}^P$ , HVK learns a per-patch reconstruction map. Each patch is compressed to a classical MPS feature vector  $\phi_{\text{MPS}}(x_p)$ , a learned linear layer sets the VQC rotation angles, and measuring a fixed set of Pauli observables yields the correlator feature map  $\Phi(x_p) \in \mathbb{R}^d$  ( $d = 27$  for HVK1D, 19 for HVK2D). With  $n = 6$  qubits, bond set  $\mathcal{E}$ , and  $c$  measured pair channels per bond,

$$d = 2n + c|\mathcal{E}| = \begin{cases} 2(6) + 3(5) = 27, & \text{HVK1D (ZZ, XX, YY on a 5-bond chain),} \\ 2(6) + 1(7) = 19, & \text{HVK2D (ZZ on 7 grid edges),} \\ 2(6) + 1(5) = 17, & \text{ZZ-only ablation control.} \end{cases} \quad (1)$$

All reported HVK1D and HVK2D results use  $d = 27$  and  $d = 19$ . A classical decoder  $D_\theta$  maps  $\Phi(x_p)$ , together with a Fourier positional encoding  $\pi(p)$ , back to a reconstructed patch  $\hat{x}_p = D_\theta(\Phi(x_p), \pi(p))$ . Training minimises the pixel-reconstruction loss with the graph-Hamiltonian energy term  $E(\cdot)$  as an auxiliary regulariser,

$$\mathcal{L}(\theta) = \sum_{p=1}^P \|x_p - \hat{x}_p\|_2^2 + \lambda E(\Phi(x_p)), \quad (2)$$

where  $\lambda$  weights the diagnostic energy term (Section 5.1 reports its effect). The shared pipeline is

$$\begin{aligned} \text{Image} &\rightarrow \{\text{Patches}\} \rightarrow \text{MPS} \\ &\rightarrow 46\text{-D}/22\text{-D Feature} \rightarrow 6\text{-D Vector} \\ &\rightarrow 27\text{-D}/19\text{-D Latent} \rightarrow \text{Decoder} \\ &\rightarrow \text{Reconstruction.} \end{aligned} \quad (3)$$

### ***Why MPS features.***

Treated as an amplitude-encoded quantum state, a patch of  $n$  sites has a coefficient tensor with  $2^n$  entries. Matrix-product-state factorisation instead represents it as a chain of small tensors bounded by a fixed bond dimension  $\chi$ . This is a compact, singular-value-controlled compression whose entanglement-entropy profile is already known to track image compressibility (Latorre 2005; Stoudenmire and Schwab 2016), and whose local expectations and correlations are cheap classical features that do not require a full state-preparation circuit.

**Table 1** The two executed configurations. Both use six qubits and the same decoder; they differ in patch size and in the measured observable set.

|                           | HVK1D               | HVK2D                   |
|---------------------------|---------------------|-------------------------|
| Benchmark                 | <i>Monalisa</i>     | multi-dataset           |
| Image size                | $256 \times 256$    | $32 \times 32$ (native) |
| Patch size                | $64 \times 64$      | $8 \times 8$            |
| MPS sites                 | 12                  | 6                       |
| Local $X/Z$ expectations  | 24                  | 12                      |
| Nearest-neighbour $ZZ$    | 11                  | 5                       |
| Bipartite entropies       | 11                  | 5                       |
| MPS feature vector        | 46-D                | 22-D                    |
| Measured observables      | 27-D                | 19-D                    |
| local $Z$ , local $X$     | 6, 6                | 6, 6                    |
| pair terms                | 5 each $ZZ, XX, YY$ | 7 grid-edge $ZZ$        |
| Fourier positional vector | 8-D                 | 4-D                     |

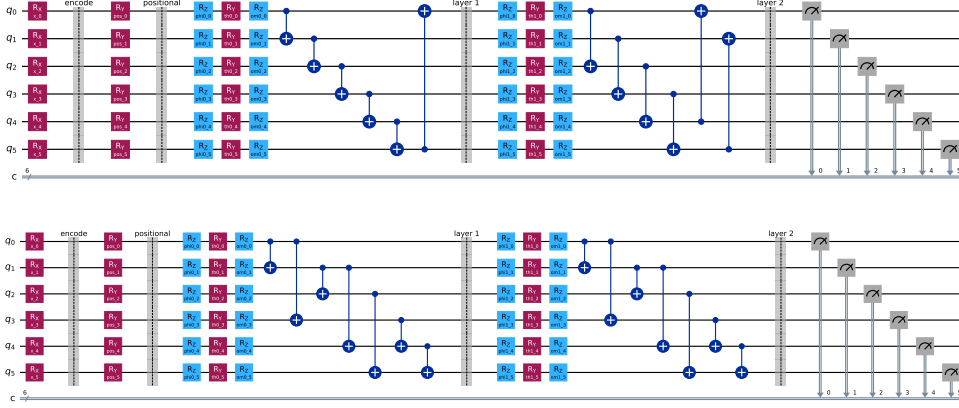
***Why a Pauli-correlator latent space rather than a raw quantum state.***

Handing the classical decoder a fixed vector of measured Pauli expectation values, instead of full state tomography, keeps the quantum-to-classical interface at  $O(\text{qubits})$  rather than exponential readout cost. It also follows the measured-observable-as-feature pattern used by quantum-enhanced feature maps generally (Havlíček et al. 2019). This choice is what makes the interpretability diagnostics of Section 5.1 and the training-dynamics diagnostics of the companion Supplementary Material possible, since a full state vector would not expose a comparably compact, physically meaningful summary.

***Why a grid rather than a chain interaction topology for HVK2D.***

Aligning the six-qubit interaction graph with the  $2 \times 3$  patch-neighbour adjacency, rather than an arbitrary linear chain, is what makes the  $D_4$  group-pooling construction of Section 4.7 well defined in the first place. The pooling operator averages over the spatial symmetries of the square, which requires a latent structure that itself respects the patch grid’s two-dimensional adjacency. HVK1D’s chain topology serves the complementary role in this design: it is the ungridded control against which the grid’s contribution is measured (Section 5.3).

Two preprocessing configurations were executed, one per topology. Both follow the same three stages. Each image is first divided into sixteen non-overlapping patches. Every patch is then  $\ell_2$ -normalised and sequentially SVD-compressed to a matrix product state of bond dimension  $\chi = 4$ . Finally, the local expectations, nearest-neighbour correlations and bipartite entropies of that state are read off as the classical MPS feature vector. The two configurations differ only in the sizes involved, which Table 1 lists in full. The remaining stages are common to both. A learned linear layer compresses the MPS feature vector to six circuit angles, one per qubit. After the circuit



**Fig. 1** The HVK1D (top) and HVK2D (bottom) variational ansätze, exactly as executed (encoding  $R_X$  rotations, positional  $R_Y$  rotations, then two entangling layers of  $R_Z$ - $R_Y$ - $R_Z$  single-qubit rotations with a CNOT ring for HVK1D or fixed grid-edge CNOTs for HVK2D). Both circuits were rebuilt gate-for-gate in Qiskit and validated against the original PennyLane training circuit before use in the hardware pilot of Section 4.4.

**Table 2** HVK variants tested in this paper.

| Variant             | Topology                  | Latent dim. | Symmetry                   |
|---------------------|---------------------------|-------------|----------------------------|
| HVK1D               | 6-qubit chain             | 27-D        | None (control)             |
| HVK2D               | $2 \times 3$ grid         | 19-D        | $D_4$ -motivated only      |
| Symmetric HVK1D     | 6-qubit chain             | 27-D        | U(1) coupling              |
| $D_4$ -pooled HVK2D | $2 \times 3$ grid, pooled | 19-D        | $D_4$ -equivariant (exact) |

is executed, a two-hidden-layer MLP (128 and 256 units) maps the measured observables together with the Fourier positional vector back to a single greyscale patch. The positional encoding follows established use as a spatial inductive bias in classical generative vision models (Xu et al. 2021). Figure 1 shows the exact HVK1D and HVK2D ansatz circuits. These were rebuilt in Qiskit for the hardware pilot of Section 4.4, and numerically verified against the training-time PennyLane circuit before use. The complete Hamiltonian, loss, positional-encoding and measurement definitions are retained in the companion supplement.

Table 2 summarises the tested variants. HVK1D uses a linear 6-qubit chain (5 nearest-neighbour bonds, 27-D observable vector); HVK2D arranges the same 6 qubits on a  $2 \times 3$  lattice matching patch-boundary adjacency (19-D latent signature); a symmetric HVK1D variant adds a U(1)-preserving coupling constraint; and a  $D_4$ -pooled HVK2D variant (Section 4.7) group-averages the observable grid over the eight symmetries of the square,  $\Phi_{D_4}(x) = \frac{1}{8} \sum_{g \in D_4} P_g^{-1} \Phi(gx)$ , which is equivariant by construction. The IBM Cloud circuit probe reported in the supplement confirms both HVK1D and HVK2D compile to depth-18, 6-qubit circuits on a Heron-class basis.

**Table 3** HVK2D same-set reconstruction across six real image datasets (per-image trained, 80–100 steps, mean  $\pm$  std over sampled images).

| Dataset        | $n$ images | PSNR (dB)        | SSIM                |
|----------------|------------|------------------|---------------------|
| CIFAR-10       | 3          | $37.75 \pm 2.38$ | $0.9986 \pm 0.0003$ |
| MNIST          | 3          | $25.77 \pm 2.23$ | $0.9783 \pm 0.0116$ |
| Fashion-MNIST  | 3          | $34.55 \pm 3.05$ | $0.9980 \pm 0.0006$ |
| PathMNIST      | 3          | $36.86 \pm 2.11$ | $0.9118 \pm 0.0627$ |
| BloodMNIST     | 2          | $36.57 \pm 2.36$ | $0.9922 \pm 0.0063$ |
| PneumoniaMNIST | 2          | $41.60 \pm 0.98$ | $0.9985 \pm 0.0006$ |

### 3 Experimental Setup

All image-reconstruction results in Sections 4.1–4.4 use the single-image *Monalisa* protocol developed for HVK: a fresh model is optimised directly on its target image, with no held-out split. These results therefore measure per-image fitting and reconstruction capability. The hardware pilot replays the resulting fitted checkpoints on real devices without retraining. The symmetry and restricted-entanglement diagnostics use the separate protocols stated in their respective sections. The training-dynamics readouts of Section 4.9 are reported as an interpretability diagnostic, with their scope set by the negative control given there. Held-out generalisation and component necessity are evaluated under the resource-matched statistical protocol of the companion supplementary study (Section 4.3). All settings report MSE and  $\text{PSNR} = 20 \log_{10}(1/\sqrt{\text{MSE}})$ ; image settings additionally report SSIM. Full implementation settings and environment versions are given in the companion supplementary document.

## 4 Results

### 4.1 Reconstruction across six real image datasets

For each image, we trained a fresh HVK2D model using non-overlapping  $8 \times 8$  patches (16 patches/image, 80–100 optimisation steps). The sample covers CIFAR-10 (Krizhevsky 2009), MNIST (LeCun et al. 1998), Fashion-MNIST (Xiao et al. 2017), and the PathMNIST, BloodMNIST, and PneumoniaMNIST subsets of MedMNIST v2 (Yang et al. 2023). Table 3 reports mean PSNR between 25.8 and 41.6 dB (SSIM 0.91–1.00). This establishes that the architecture fits varied image structures under a common protocol.

### 4.2 Scope: models do not transfer zero-shot

A model optimised on one image alone does not transfer zero-shot to a structurally different image. On *Monalisa*, zero-shot transfer to a second image reaches 8.31 dB PSNR, with SSIM 0.1044. Extending training to include the second image recovers 28.63 dB, with SSIM 0.9858. The per-image results above therefore measure per-image optimisation and reconstruction capability. Section 4.3 tests the held-out case directly.

### 4.3 Competitive with resource-matched classical baselines

The results above measure per-image fitting. A separate question is whether HVK’s feature map improves reconstruction on real, *held-out* images under a resource-matched multi-seed protocol. That question is answered in the companion document, *Supplementary Material for the Hamiltonian Vision Kernel: Resource-Matched Ablations, Validation, and Reproducibility*.

On held-out CIFAR-10, the **raw/local linear control** reaches  $18.80 \pm 1.42$  dB, against HVK2D’s  $18.12 \pm 1.54$  dB. Aggregating held-out images within each split gives an HVK2D-minus-control difference of  $-0.68$  dB, with bootstrap 95% CI  $[-0.91, -0.46]$  dB and Wilcoxon  $p = 0.0625$  at  $n = 5$  seeds. The same numerical direction holds on five further real datasets. The tested HVK map is therefore informative, but it is not shown to outperform the simpler maps on ordinary held-out reconstruction.

A TOST equivalence test at a declared  $\pm 1$  dB margin confirms that this gap is small enough, in a pre-declared and data-independent sense, to call the two *competitive* rather than merely “not shown to differ”. **Five of seven distinct** tested controls are statistically equivalent to HVK2D within that margin. The two weakest controls, strict classical random features and random-VQC, are correctly excluded, because HVK2D substantially outperforms them. We report this as a complement to the positive entanglement and symmetry results below. It delineates their tested boundary and records the provenance checks that prevented exploratory artifacts from becoming claims.

A matched-budget, multi-seed rerun of the component-ablation table reaches the same conclusion on ordinary per-image reconstruction. The rerun uses 240 steps and 3 seeds per variant. **It replaces an earlier table that we withdrew, because the repository does not retain the per-seed records needed to reproduce it.** A plain classical-replacement control ties or slightly exceeds the quantum baseline, and only a circuit-free random-VQC control performs clearly worse. This rerun uses a descriptive pooled-standard-deviation comparison rule at 3 seeds rather than a calibrated test, and full results are in the companion supplement. Section 5.2 discusses why this ties-with-classical result and the entanglement-necessity result of Section 4.8 are not in tension.

### 4.4 A real hardware reconstruction pilot

The results above are noiseless-simulator results. We now report a small but complete image-reconstruction pilot executed on real IBM Quantum hardware (`ibm_fez`, a 156-qubit Heron-class processor), going beyond the compiled-circuit feasibility probe documented in the supplement.

The pilot covers five images. These are the full 16-patch *Monalisa* benchmark and four native-resolution CIFAR-10 images (*cat*, *ship* $\times 2$ , *airplane*), each optimised directly on its own target image following the same per-image protocol as every result in this paper. For each image we took an already-trained HVK1D/HVK2D checkpoint’s exact learned parameters and rebuilt its measurement circuits gate-for-gate in Qiskit. We verified the translation against the original PennyLane simulator to within numerical noise, at most  $7.9 \times 10^{-4}$  maximum absolute error per observable

**Table 4** Real IBM hardware reconstruction pilot (backend `ibm_fez`, 256 shots/basis, no retraining). PSNR compares hardware-decoded and simulator-decoded reconstructions against the same ground-truth image.

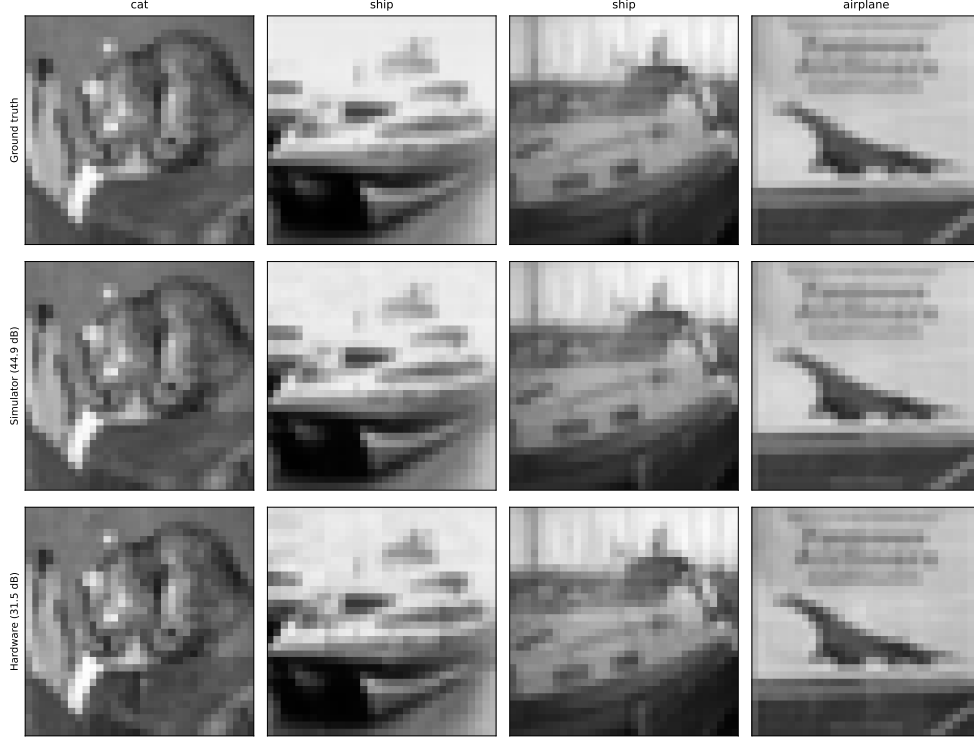
| Image                           | Patches | Sim. PSNR (dB)   | HW PSNR (dB)     |
|---------------------------------|---------|------------------|------------------|
| Monalisa (HVK1D)                | 16      | 37.99            | 25.90            |
| CIFAR cat (HVK2D)               | 16      | 44.85            | 31.52            |
| CIFAR ship, hydrofoil (HVK2D)   | 16      | 36.56            | 26.44            |
| CIFAR ship, sea boat (HVK2D)    | 16      | 42.57            | 31.20            |
| CIFAR airplane (HVK2D)          | 16      | 46.79            | 29.10            |
| CIFAR mean $\pm$ std (4 images) | —       | $42.69 \pm 3.84$ | $29.56 \pm 2.03$ |



**Fig. 2** *Monalisa* reconstructed from real IBM Quantum hardware measurements (right) against the same trained checkpoint’s noiseless-simulator replay (centre) and ground truth (left). The hardware image is visibly noisier and exhibits patch-boundary banding, while the subject remains recognisable.

across every patch, with full validation numbers in the companion supplementary document. We then executed the same circuits on real hardware with 256 shots per measurement basis. This is a fixed per-basis shot budget, rather than an adaptive shadow-tomography-style estimation scheme (Huang et al. 2020). The hardware-measured observables were decoded by the same trained classical decoder, with no retraining and no simulator involved anywhere in the reconstruction path.

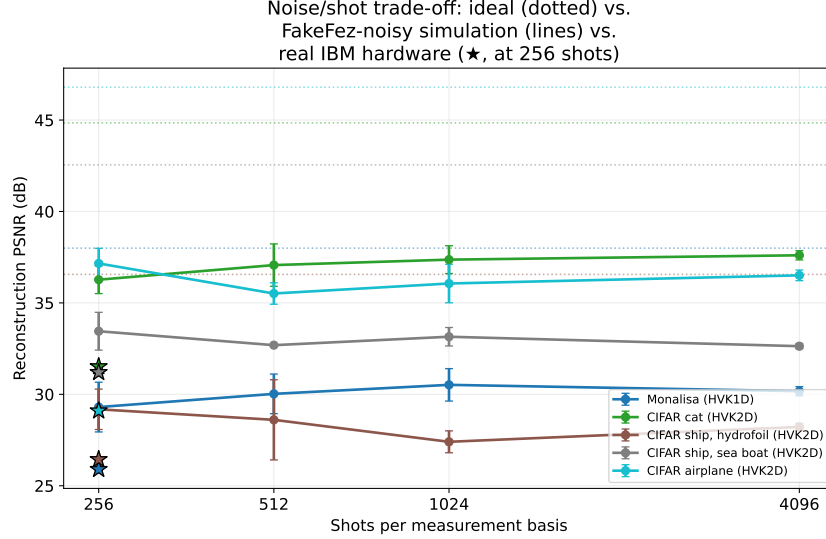
Reconstructions from real, noisy hardware measurements retain recognisable image structure. *Monalisa* reaches 25.90 dB PSNR from hardware measurements, against a 37.99 dB noiseless-simulator replay of the identical circuit (Figure 2). The four CIFAR-10 images average  $29.56 \pm 2.03$  dB on hardware against  $42.69 \pm 3.84$  dB in simulation (Figure 3, Table 4). The observed gap is consistent with noise in these unmitigated device executions, although the five-image pilot does not isolate individual noise sources. Every hardware reconstruction remains above the random-latent control range (11–15 dB in the companion supplementary study). The pilot used 176 circuits; the contemporaneous account-meter reading increased by 16 quantum-processor seconds, with the provenance limitation documented in the supplement.



**Fig. 3** Four CIFAR-10 images reconstructed from real IBM Quantum hardware measurements (bottom row) against noiseless-simulator replays of the same trained HVK2D checkpoints (middle row) and ground truth (top row).

Five images constitute a pilot, and we make no hardware-advantage claim. What the pilot does establish is that the trained HVK observable channel survives execution on present-day hardware. The learned parameters, the measurement circuits and the classical decoder all transfer unchanged, and the decoded images remain quantitatively informative at 256 shots per basis on an unmitigated device. This is the sense in which HVK runs on real quantum hardware. The companion supplementary document gives the full methodology, covering exact gate sequences, per-basis measurement construction, job IDs, and pre-submission local-validation results.

The same checkpoint-replay methodology supports a further hardware-validation exercise, reported in the companion Supplementary Material. Fresh HVK1D checkpoints were trained at  $N \in \{4, 6, 8\}$  and replayed epoch-by-epoch on real IBM Quantum hardware and on the IonQ cloud simulator (IonQ 2026). That exercise serves the training-dynamics diagnostic of Section 5.1. **It stays out of the main text because it is a single-patch, single-seed probe, and because the threshold it tracks fires equally with the Hamiltonian term present and absent (Section 4.9). We therefore treat it as an interpretability readout, on the same footing as the traces it supports.**



**Fig. 4** Calibrated device-noise simulation (FakeFez, mean  $\pm$  std over three executions). Dotted lines show noiseless ceilings and stars show real-hardware measurements. Increasing 256 to 4096 shots changes mean PSNR by only  $-0.05$  dB.

#### 4.5 Noise/shot trade-off: a free simulator complement

The five-image pilot above used a single shot budget (256/basis) and one execution per image, which bounds how much it can say about noise sensitivity on its own. Real IBM Quantum time is a scarce free-tier resource, so we separate what a *calibrated device-noise simulation* can establish at zero hardware cost from what genuinely requires hardware. Using `qiskit-aer` with the live calibration snapshot of `ibm_fez` (FakeFez, no quantum-processor time consumed), we replay the same already-trained checkpoints’ circuits at four shot budgets (256/512/1024/4096 shots per basis, three repeated executions each) and compare against both the noiseless-simulator ceiling and the real-hardware point already reported above.

Two findings follow, averaged over the five checkpoints. First, shot noise accounts for only part of the noiseless-to-real gap. Moving from the noiseless ceiling to a calibrated-noise simulation at the same 256-shot budget already costs 8.68 dB on average, before any real device is involved. Second, more shots does not reliably recover it. Increasing to 4096 shots per basis changes PSNR by only  $-0.05$  dB on average across the five checkpoints, with individual images ranging from a 1.33 dB gain to a 0.96 dB loss. At this circuit size and shot range, shot count is not a lever that reliably improves reconstruction quality.

Comparing the 256-shot noisy simulation directly against the real-hardware point isolates what the calibration snapshot does *not* capture. That comparison leaves a further 4.24 dB average gap, ranging from 2.25 to 8.06 dB across images. We attribute this residual to drift between calibration and execution, crosstalk, and correlated errors. The airplane image is the clear outlier, with an 8.06 dB residual gap, roughly double any other image. We report it rather than exclude it, since a single-execution,

**Table 5** Real-hardware anchor points: same checkpoints, no retraining, on two further backends. The original pilot’s `ibm_fez` point is shown for reference.

| Image     | Backend                         | Shots | PSNR (dB) |
|-----------|---------------------------------|-------|-----------|
| Monalisa  | <code>ibm_fez</code> (original) | 256   | 25.90     |
| Monalisa  | <code>ibm_marrakesh</code>      | 256   | 25.94     |
| Monalisa  | <code>ibm_kingston</code>       | 1024  | 26.10     |
| CIFAR cat | <code>ibm_fez</code> (original) | 256   | 31.52     |
| CIFAR cat | <code>ibm_kingston</code>       | 256   | 28.93     |
| CIFAR cat | <code>ibm_kingston</code>       | 1024  | 31.24     |

five-image pilot cannot distinguish an image-specific effect from ordinary job-to-job variation. This calibrated-noise simulation is a free, repeatable complement to the pilot, not a substitute for it. It bounds how much of the observed hardware gap is attributable to modelled device noise rather than unmodelled effects, using zero additional quantum-processor time.

#### 4.6 Real-hardware anchor points: repeated execution and a second backend

The simulated trade-off above is bounded, in turn, by whether it actually reflects real devices. We confirmed this directly on hardware, using a small, quota-tracked slice of the free-tier IBM Quantum allowance of four jobs, well under 1% of the monthly limit. Two checkpoints, *Monalisa* and *CIFAR cat*, were replayed with no retraining on `ibm_marrakesh` and `ibm_kingston`. Both backends are distinct from the original pilot’s `ibm_fez`. We ran them at the original 256-shot budget and, separately, at 1024 shots.

*Monalisa* reproduces almost exactly across backends at matched shots (25.94 vs. 25.90 dB, `ibm_marrakesh` vs. `ibm_fez`), and gains only 0.16 dB from quadrupling shots to 1024, consistent with the simulated finding above that shot count is not a reliable lever. The *CIFAR cat* point is the more informative one. On `ibm_kingston` at the original 256-shot budget it reaches 28.93 dB, which is 2.59 dB below the `ibm_fez` pilot point. Its own 1024-shot execution on the same backend recovers to 31.24 dB, within 0.28 dB of the original. This is real job-to-job and backend-to-backend variation of a magnitude comparable to the unmodelled residual identified by the simulator comparison (Section 4.5). It provides direct evidence, not merely a simulated bound, that a single 256-shot execution on a single backend is not yet a stable estimate of this pipeline’s hardware performance, and that repeated execution matters at least as much as backend choice or shot count individually.

**Table 6**  $D_4$  equivariance error for unpooled and pooled observable-grid maps (lower is better; the pooled map is group-averaged over all eight square symmetries).

| Model                  | Mean error             | Std. error             |
|------------------------|------------------------|------------------------|
| $D_4$ -pooled HVK2D    | $9.57 \times 10^{-17}$ | $1.26 \times 10^{-17}$ |
| No positional encoding | $7.40 \times 10^{-1}$  | $1.52 \times 10^{-1}$  |
| HVK2D positional       | $8.15 \times 10^{-1}$  | $1.29 \times 10^{-1}$  |
| Local/raw              | $8.37 \times 10^{-1}$  | $1.31 \times 10^{-1}$  |

#### 4.7 Exact $D_4$ -equivariant observable pooling

A  $D_4$ -pooled HVK2D observable-grid map is equivariant to the eight symmetries of the square *by construction*:

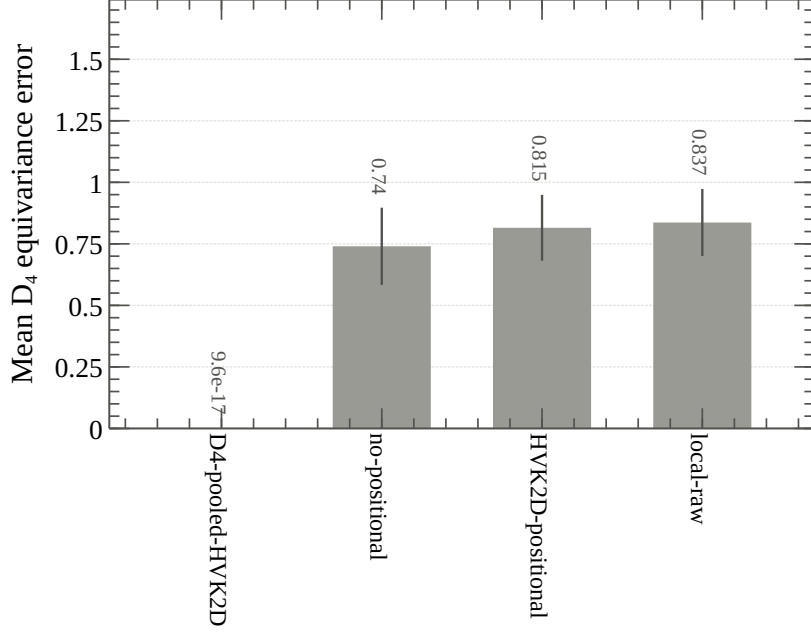
$$\Phi_{D_4}(x) = \frac{1}{8} \sum_{g \in D_4} P_g^{-1} \Phi(gx), \quad \Phi_{D_4}(hx) = P_h \Phi_{D_4}(x) \quad \forall h \in D_4, \quad (4)$$

up to floating-point error. We verify this numerically over 7000 patch transforms of cached CIFAR images (Table 6, Figure 5). The pooled map’s equivariance error is  $9.57 \times 10^{-17} \pm 1.26 \times 10^{-17}$ , which is floating-point noise. The unpooled positional map has an error of  $0.815 \pm 0.129$ , and the local/raw control has  $0.837 \pm 0.131$ . Both are of order one. The patch grid’s geometry therefore makes the unpooled map  $D_4$ -*motivated* only, and the explicit pooling construction is what delivers the property. This construction operates at the feature-grid level and serves here as a numerical diagnostic, rather than as a component of the trainable reconstruction pipeline. Integrating it end-to-end (Section 5.4) is concrete future work.

#### 4.8 Entanglement necessity in the nonlocal-structure protocol

The diagnostic tests whether an entangling pair-observable channel is needed to fit a target that depends on pair correlations. The target is a fixed synthetic function of *simulated* patch features, and those features are not concatenated into any candidate’s own input block. The companion supplement gives the leakage-audit procedure that rules out circularity. Every control receives the same raw inputs as HVK2D. **The task is by construction favourable to pair-observable structure, which is why we read it as a representational statement and not as a performance comparison.**

Table 7 and Figure 6 report the result across five seeds. HVK2D’s entangling observable channel reaches mean  $R^2 = 0.9735$ , or 32.77 dB PSNR. This is a 17.74 dB improvement over a no-entanglement control ( $R^2 = 0.019$ ) and 18.52 dB over random-VQC latents ( $R^2 = -0.173$ ). Freezing the quantum circuit, so that features are fixed and only the readout is trained, still reaches  $R^2 = 0.853$ . A parameter-matched classical control ( $R^2 = -0.030$ ) and a frozen-classical control ( $R^2 = -0.903$ ) fail outright. **The raw-linear classical control ( $R^2 = 0.0133$ ) is in the table but is**



**Fig. 5**  $D_4$  equivariance diagnostic. The unpooled positional map is not equivariant; the group-pooled map is equivariant to numerical precision.

**not named here; whether it should be depends on the duplicate-control question raised separately.**

The mechanism is explicit under the fixed linear readout. A linear map of single-patch expectations  $\{\langle O_i \rangle\}$  cannot form the cross-term  $\langle O_i \rangle \langle O_j \rangle$  that the target depends on. The entangling measurement supplies the joint correlator  $\langle O_i O_j \rangle$  directly as a feature. An entangling pair-observable channel is therefore *representationally necessary within the tested feature class and fixed linear-readout protocol* to fit this target. A sufficiently nonlinear classical readout could in principle recover the same cross-terms, which is why we scope the claim to the linear protocol rather than to classical models in general.

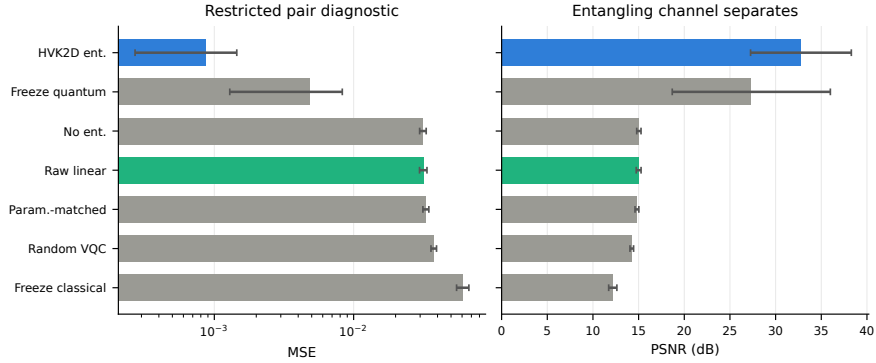
#### 4.9 A scoped training-dynamics diagnostic

Beyond the results above, we tracked two exploratory training-dynamics readouts. These are a magnetisation-style statistic  $M_z(t)$  and a signed energy-to-entanglement ratio  $R_{ES}(t)$ . Both drift smoothly over training with run-to-run fluctuations, and neither shows a sharp transition. Any apparent onset is a within-run median-plus- $2\sigma$  threshold crossing, rather than a calibrated statistical test.

A negative control fixes the scope. We applied the identical protocol to a classical-replacement variant, in which the Hamiltonian energy is identically zero throughout training. That variant crosses the threshold in all 4/4 tested cases, which is as often as the Hamiltonian-regularised model’s 4/4. The threshold therefore fires whether the

**Table 7** HVK2D restricted pair-correlation diagnostic across five seeds. Deliberately favourable to pair-observable structure; evidence for entanglement-necessity under a matched linear readout.

| Model                        | Mean MSE                | Mean PSNR (dB) | Mean $R^2$ |
|------------------------------|-------------------------|----------------|------------|
| HVK2D entangling observables | $8.5969 \times 10^{-4}$ | 32.77          | 0.9735     |
| Freeze quantum               | $4.7904 \times 10^{-3}$ | 27.34          | 0.8533     |
| No entanglement              | $3.1461 \times 10^{-2}$ | 15.03          | 0.0191     |
| Raw-linear classical         | $3.1654 \times 10^{-2}$ | 15.00          | 0.0133     |
| Parameter-matched classical  | $3.3033 \times 10^{-2}$ | 14.82          | -0.0297    |
| Random VQC                   | $3.7616 \times 10^{-2}$ | 14.25          | -0.1732    |
| Freeze classical             | $6.0973 \times 10^{-2}$ | 12.17          | -0.9025    |



**Fig. 6** Restricted pair-correlation diagnostic summary (mean  $\pm$  std over five seeds). The entangling observable feature map dominates every control on this deliberately favourable task.

Hamiltonian term is present or absent, and it does not track that term. We make no transition claim, and report these traces as an interpretability readout on the learned energy and entanglement. The companion Supplementary Material gives the full protocol and tables.

## 5 Discussion

### 5.1 Why the physics-informed regulariser was inert

Table 8 shows the Hamiltonian energy term is a mild drag on reconstruction quality. The legacy objective adds a signed mean energy to the reconstruction loss. Because learned energy can go negative, the optimiser can lower total loss by driving energy down without improving reconstruction. Removing the term improves PSNR from 38.63 to 44.56 dB. A contrastive variant, assigning lower energy to correct feature-position pairs than to shuffled pairs, recovers part of the gap (42.92 dB) but stays below its no-energy counterpart. These are fresh same-environment measurements

**Table 8** Hamiltonian-sector controls (single-seed, fresh same-environment reproduction: *Monalisa*, 200 steps, seed 42). The energy objective is diagnostically useful but does not improve reconstruction accuracy. See the Supplementary Material for the full reproducibility investigation and additional observable-sector controls.

| Experiment                     | PSNR (dB) | Interpretation                                                             |
|--------------------------------|-----------|----------------------------------------------------------------------------|
| Legacy baseline, signed energy | 38.63     | Standard energy-regularised run                                            |
| No energy loss                 | 44.56     | Removing energy improves PSNR                                              |
| Contrastive Hamiltonian core   | 42.92     | Stronger objective helps modestly vs. baseline, still below no-energy-loss |

(*Monalisa*, 200 steps, seed 42), and they supersede earlier values produced by a script that cannot be audited.

The learned couplings enter only the energy term and never reach the decoder. The Hamiltonian is therefore a *read-only* observable of the trained representation rather than an inductive bias on reconstruction, and this is the sense in which we call it a diagnostic. A physically motivated regulariser has room to help when it resolves an otherwise underdetermined objective. This patch-level pixel task does not meet that condition, because the reconstruction loss already dominates the optimisation signal at this model scale. The Supplementary Material gives the full account.

The training-dynamics readouts of Section 4.9 provide a descriptive window into how the learned energy and entanglement evolve. They are descriptive readouts rather than calibrated statistical tests, and the thermodynamic reading is outside their scope.

## 5.2 Task-dependent representational scope

Whether a quantum feature map offers any advantage over a classical model is a property of the specific data and target, not a fixed property of the model class. Dequantization results have repeatedly shown that apparent quantum advantages on structured, low-entanglement classical data can be matched once the right classical feature is identified (Huang et al. 2021; Tang 2019). Where provable separations do exist, they are established for carefully constructed learning problems rather than for generic structured data (Gyurik and Dunjko 2023).

The restricted diagnostic of Section 4.8 is constructed so that the relevant structure is a genuine distant-element product, which raw or local features cannot represent under a fixed linear readout. That is the regime in which HVK’s entangling channel separates. Natural image patches at the resolution and patch size used here are low-entanglement, locally structured data, where a linear or shallow classical readout is already close to sufficient. The distinction lies in the measured pair-observable channel rather than in the circuit. The ansatz uses strongly-entangling layers and does generate an entangled state, but whether the measured channel carries task-relevant information is a property of the target. On natural-image targets the pair observables add little beyond the local ones. On the constructed nonlocal target they are necessary. Together with the held-out comparisons of Section 4.3, these results delineate where the tested entangling representation does and does not yield a measurable benefit.

HVK’s three ingredients each recur individually in the hybrid quantum-vision literature, covering MPS compression of structured input (Stoudenmire and Schwab 2016; Han et al. 2018; Cirac et al. 2021; Guala et al. 2023), a Pauli-correlator latent space handed to a classical decoder (Havlíček et al. 2019), and a physically motivated auxiliary regulariser alongside a task loss (Shi et al. 2023). Its specific instantiation, a six-qubit VQC with a 19–27-D correlator vector and a graph-Hamiltonian term, sits within the parameter ranges typical of that literature. The measured behaviour is therefore evidence about the design pattern itself. We do not claim the boundary transfers unchanged to every dataset, patch size, or qubit count.

### 5.3 Topology alignment as an inductive bias

Aligning the latent interaction graph with image-patch adjacency (HVK2D vs. HVK1D) is an architectural inductive bias, and the companion supplementary study tests it directly under matched held-out conditions. Its real-circuit subset (3 seeds, 3 held-out CIFAR-10 images) gives an HVK1D-minus-HVK2D mean PSNR difference of 0.16 dB. After averaging images within seed, the bootstrap 95% CI is  $[-0.38, 0.46]$  dB and the seed-level Wilcoxon test gives  $p = 0.50$ . A broader five-seed surrogate comparison across CIFAR-10, Fashion-MNIST, and PathMNIST likewise finds numerically small topology differences, at most 0.07 dB in mean PSNR. HVK2D completes the matched 90-step runs in roughly half the wall-clock time. At the tested scale, topology is therefore primarily a resource-cost choice rather than a reliable accuracy advantage. These tests do not establish formal equivalence.

Table 9 contrasts HVK’s structural properties against reference frameworks, including the three nearest hybrid-vision architecture families. Its distinguishing features are a Pauli-correlator latent space, a 2D Fourier positional bias with enforced  $D_4$  pooling, and a Hamiltonian energy diagnostic, each documented and scoped in the preceding sections.

### 5.4 Validated scope

- Every reconstruction result in this paper (Sections 4.1–4.2) is per-image trained. The companion supplementary study’s held-out protocol is the appropriate evidence for a generalisation claim.
- The  $D_4$ -pooled map (Section 4.7) is a feature-grid-level construction and numerical diagnostic. Integrating it into the trainable end-to-end pipeline is future work (Section 6).
- The entanglement-necessity result (Section 4.8) is scoped to a task constructed to require nonlocal structure. Ordinary natural-image reconstruction lies outside that scope (see Section 5.2).
- The hardware pilot (Section 4.4) is a five-image, single-trial, unmitigated pilot rather than a statistically resolved hardware benchmark.
- The training-dynamics traces (Section 4.9) are an interpretability readout. The threshold fires whether the Hamiltonian term is present or absent, at 4/4 in both cases, with the full table in the Supplementary Material.

**Table 9** Architectural differentiation from representative baseline designs, including the three nearest hybrid-vision architecture families (register-based QAE, quantum-kernel vision, TN-circuit classifier).

| Axis                    | Conv. AE baseline           | GAN baseline              | Register-based QAE        | Quantum-kernel vision    | TN-circuit classifier   | HVK (ours)                   |
|-------------------------|-----------------------------|---------------------------|---------------------------|--------------------------|-------------------------|------------------------------|
| Latent representation   | Continuous $\mathbb{R}^d$   | Continuous $\mathbb{R}^d$ | Quantum register          | Scalar kernel value      | MPS/circuit state       | Pauli-correlator vector      |
| Spatial bias            | Convolutional               | Convolutional             | Encoding-dependent        | Encoding-dependent       | MPS chain ordering      | 2D Fourier encoding          |
| Auxiliary objective     | Bottleneck / reconstruction | Adversarial loss          | Fidelity / reconstruction | None (kernel value only) | Classification loss     | Hamiltonian diagnostic       |
| Enforced group symmetry | Not in baseline             | Not in baseline           | Not in reference design   | Not in reference design  | Not in reference design | $D_4$ observable pooling     |
| Evaluated role          | Reconstruction              | Reconstruction            | Compression               | Classification           | Classification          | Reconstruction + diagnostics |

- The matched real-circuit topology comparison uses 3 seeds and 3 held-out images per topology, and the broader five-seed comparison is surrogate-based. These results support a resource-cost interpretation rather than formal equivalence of the topologies.
- Every result here is at a six-qubit, depth-18 scale. Trainability of variational models is known to degrade with width and depth through barren-plateau effects (McClean et al. 2018; Larocca et al. 2025), so whether the tested boundary moves at larger scale is an open question.

## 6 Conclusion

We presented the Hamiltonian Vision Kernel, a hybrid quantum–classical architecture for image reconstruction that assembles matrix-product-state patch features, a measured Pauli-correlator latent space, enforced  $D_4$ -equivariant observable pooling, and a learnable Hamiltonian energy diagnostic into a single pipeline (Section 5.2). Across six real image datasets, per-image-trained HVK2D models reconstruct with high fidelity. On a resource-matched, multi-seed held-out comparison, a pre-declared TOST equivalence test shows HVK2D is *competitive with*, rather than inferior or superior to, simple classical controls (Section 4.3). A hardware pilot replayed trained checkpoints on IBM Quantum devices without retraining, establishing feasibility at a five-image, single-trial scope (Section 4.4).

HVK’s case rests on capabilities a classical reconstruction pipeline does not provide. Pair observables are necessary on a task requiring nonlocal structure, within the tested fixed-readout feature class (Section 4.8). The Hamiltonian energy term serves

as an interpretable diagnostic of the trained representation (Section 5.1). The pipeline executes natively on real quantum hardware. It is also exactly  $D_4$ -equivariant by construction, which we report as a design-correctness property rather than a quantum capability, since the same Reynolds averaging applied to a classical feature map is equally exact. We make no claim of quantum advantage. HVK is, at this scope, a new architecture that works, is competitive with resource-matched classical baselines, and runs on real quantum hardware.

Future work should integrate the  $D_4$ -pooled map into an end-to-end trainable pipeline (Section 5.4), scale the hardware pilot toward a statistically resolved benchmark with error mitigation and repeated trials, search for further real-world tasks with nonlocal structure where HVK’s entanglement-sensitive separation translates into a reconstruction advantage, and extend the resource-matched, leakage-audited evaluation protocol to other hybrid quantum vision architectures.

**Supplementary information.** A companion document, *Supplementary Material for the Hamiltonian Vision Kernel: Resource-Matched Ablations, Validation, and Reproducibility*, accompanies this manuscript.

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## Statements and Declarations

*Competing interests.* The authors declare no competing interests.

*Funding.* “No funding was received for conducting this study.”

*Data availability.* Dataset provenance and access routes are cited in the manuscript and detailed in the supplementary artifact map. All datasets used (CIFAR-10, MNIST, Fashion-MNIST, and the PathMNIST, BloodMNIST and PneumoniaMNIST subsets of MedMNIST v2) are publicly available.

*Code availability.* The source code, experiment drivers, machine-readable result summaries, validation arrays, and figure inputs supporting this study are included in the accompanying repository. The file `project_artifacts/submission_claim_audit.json` maps each numerical headline claim to its retained source artifact, covering the resource-matched held-out comparison, on which the tested HVK map is competitive with rather than superior to the classical controls, and the withdrawn training-dynamics material. Automated tests verify those mappings. IBM Quantum job identifiers and reconstructed outputs are retained; as disclosed in the supplement, the raw account-usage service response was not serialised, so the reported quota change is an execution-log record rather than a repository-recomputable measurement.

*Author contributions.* **Spارشو Chakraborty:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Siddhartha Patra:** Conceptualization, Supervision, Writing – review & editing, Resources. Both authors reviewed and approved the final manuscript.

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